Position-Aware Lower Witnesses for a Literal Prime-Shell Mask Defect

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Abstract

We study the literal masked prime-shell matrix isolated in TPC-347. Writing the physical interval block as an unmasked convolution compression plus a divisibility-mask defect, we derive an exact coordinate formula for every defect column. If J_I is the set of interval positions divisible by at least one active shell prime, then the induced Euclidean norm obeys the exact finite-dimensional lower bound

$$\|D_I\|_{2\to 2} \geq W_I(D) := \max_{t\in J_I} \|D_I e_t\|_2.$$

This witness uses only declared mask-hit positions and does not fit a leading eigenvector. On a locked grid of 192 rows, all rows have a positive witness; the best-hit witness is 0.453958762219–0.897148966365 of the defect spectral norm and 0.0183057714619–0.336311065586 of the unmasked ideal norm. These are finite observations, not growing lower bounds. The result therefore sharpens a scoped obstruction to deleting the masks while leaving the source-uniform arithmetic L^2 problem open.

1 Question and scope

The twin-prime route currently contains a literal finite prime-shell operator with two endpoint divisibility restrictions. The preceding TPC-347 release showed that replacing this physical object by its unmasked convolution can hide a sizeable finite defect. The present paper asks a narrower question: can that defect be witnessed by a deterministic position-aware vector, with no use of its leading eigenvector and no sign heuristic?

Our answer is yes in a precise finite sense. The main theorem is elementary operator-norm linear algebra, but its use here is structural: it identifies the positions at which the literal masks force a nonzero column and supplies a reproducible lower-witness audit. Nothing below is an asymptotic estimate, a source-uniform arithmetic theorem, or a proof of the twin-prime conjecture.

2 The literal object

Let $I = \{o, o+1, \dots, o+M-1\}$, let R_I denote restriction, and let E_I denote zero extension. For a shell prime p, define the diagonal projection

$$(P_p f)(n) = \mathbf{1}_{p\nmid n} f(n).$$

The translated residue kernel K_p has the zero diagonal and the same height and exponent normalisation as TPC-347. With the declared sign $\varepsilon_p \in \{\pm 1\}$, the physical and unmasked blocks are

$$A_I = \sum_p \varepsilon_p R_I P_p K_p P_p E_I,$$

$$T_I = R_I \left(\sum_p \varepsilon_p K_p \right) E_I, \qquad D_I = A_I - T_I.$$

The distinction between A_I and T_I is essential: T_I is an interval compression of a translation-invariant convolution, whereas A_I retains the absolute-position masks.

For one prime, the projection algebra gives

$$P_p K_p P_p - K_p = (P_p - I)K_p P_p + K_p(P_p - I). \tag{1}$$

Thus

$$D_I = \sum_p \varepsilon_p R_I \big((P_p - I)K_p P_p + K_p(P_p - I)\big) E_I. \tag{2}$$

3 Exact position formula and lower witness

Write e_t for the unit coordinate vector associated with $t \in I$. Applying (1) gives two cases. If $p \mid t$, then $P_p e_t = 0$ and the right-mask term is $-K_p e_t$. If $p \nmid t$, then $(P_p - I)e_t = 0$ and the left-mask term retains only output coordinates divisible by p. Consequently

$$D_I e_t = -\sum_{p \mid t} \varepsilon_p R_I K_p e_t + \sum_{p \nmid t} \varepsilon_p R_I (P_p - I) K_p e_t. \tag{3}$$

This identity retains both sides of the physical mask; it makes no assertion that the prime contributions have a common sign.

Define the mask-hit set and its coordinate envelope by

$$J_I = \{t \in I : \text{there is an active shell prime } p \text{ with } p \mid t\}, \qquad W_I(D) = \max_{t \in J_I} \|D_I e_t\|_2.$$

Theorem 1 (coordinate lower witness)**.** *For every finite matrix D_I and every nonempty J_I,*

$$\|D_I\|_{2\to 2} \geq W_I(D). \tag{4}$$

In particular, the bound applies to the defect in (2) without any positivity, symmetry, or cancellation assumption.

Proof. Each e_t is a unit vector. With the standard induced-norm convention [1],

$$\|D_I\|_{2\to 2} = \sup_{\|x\|_2 = 1} \|D_I x\|_2 \geq \|D_I e_t\|_2.$$

Taking the maximum over $t \in J_I$ proves (4). □

Remark 1. *The selector is not an eigenvector optimization: J_I is fixed by the declared interval and shell, and the only optimization is the explicit maximum over its coordinate columns. We also record the first-hit coordinate as a non-adaptive control.*

4 Frozen audit protocol

The producer locks the TPC-347 code and canonical certificate. We retain its physical object and use

$$o \in \{40097, 48097\}, \quad M \in \{256, 512, 1024\}, \quad Q \in \{24, 36, 54, 80\}, \quad s \in \{1, 2\},$$

with $H = 66$. The four predeclared sign laws are `all_plus`, `alternating_index`, `mod4_character`, and `half_split`. This gives $2 \cdot 3 \cdot 4 \cdot 2 \cdot 4 = 192$ rows.

For each row we reconstruct A_I, T_I, and D_I, form J_I, and store the first-hit and best-hit column norms. The independent checker reverses the shell accumulation order and does not import the producer. It recomputes all 192 rows, the two-sided formula (3), and a rational six-point anchor. A separate stress suite mutates the selector, each side of the mask formula, the anchor, the census, and the claim firewall.

Table 1: TPC-348 position-aware witness audit.

Quantity	Certified finite readout
Rows	192
Rows with positive mask-hit witness	192/192
Mask-hit count per row	30–169
Formula records	192
Maximum formula discrepancy	$2.0872192863 \times 10^{-14}$
$W_I(D)/\|D_I\|$	0.453958762219–0.897148966365
$W_I(D)/\|T_I\|$	0.0183057714619–0.336311065586
First-hit $\|D_I e_t\|/\|D_I\|$	0.188855872493–0.533179477634

5 Finite results

Table 1 gives the complete aggregate readout. The quantity $\|T_I\|$ is the norm of the unmasked ideal comparison, not the physical operator. The ratios are reported only to describe the finite panel.

The first-hit interval is useful as a hostile control: it is chosen by the smallest mask-hit position rather than by the largest observed column. The best-hit envelope is nevertheless fully declared, and the same selector is recomputed independently. In particular, every row has a positive witness, but the minimum ratio to the ideal norm is modest; no fixed power of the main variable can be inferred.

5.1 Exact rational anchor

For $I = \{1, \ldots, 6\}$, $Q = 4$, $s = 1$, and all-plus signs, the shell is $\{5, 7\}$ and $J_I = \{5\}$. The selected fifth defect column has exact squared norm

$$\|D_I e_5\|_2^2 = \frac{126400483271766338965333 3}{162252681195863096059456}.$$

The machine certificate stores the six rational column entries and a canonical digest; the independent replay reproduces both the selector and the digest.

6 Adversarial checks and claim boundary

The normal and optimized producer outputs agree byte-for-byte. The reverse shell independent replay and both normal/optimized stress runs pass. The stress mutations reject a wrong coordinate, an omitted left mask, an omitted right mask, a changed exact anchor, a changed census, and a false Gate-B flag. These are integrity checks on the computational package, not replacements for the theorem.

The logically strongest statements in this release are:

- the two-sided projection expansion and the position formula are exact finite identities;
- the coordinate lower-witness inequality is exact induced-norm linear algebra;
- the 192-row positive-witness and formula census is numerically certified for the declared panel;
- mask deletion is refuted only as a uniformly negligible operation on that finite panel.

We do *not* claim a source-uniform arithmetic L^2 estimate, a uniform bound for the masked operator, a growing defect lower bound, a fixed-power payment, Route-B reassembly, or a twin-prime conclusion. The official Session-named Route-A/Route-B evaluator files are absent from this checkout; the local bridge is explicitly fail-closed.

7 Conclusion and next question

TPC-348 turns the mask-defect observation into a reusable finite interface: literal block $\to$ two-sided projection defect $\to$ mask-hit set $\to$ coordinate lower witness. The result shows that a position-aware defect cannot be dismissed by an unstructured finite remainder argument on the declared grid. The next minimal question is whether a prime-balanced signed combination of hit coordinates produces additional structure that survives the same hostile controls. Until such a result exists, the source-native arithmetic L^2 gate remains open.

References

[1] John B. Conway. *A Course in Functional Analysis*. Springer, 2 edition, 2000.